



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

WEB TECHNOLOGY (SECJ3483)

SEM 2 (2025/2026)

INDIVIDUAL ASSIGNMENT

Web Application & Robotic Process Automation (RPA)

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DEMO VIDEO LINK:

<https://youtu.be/-SuaH2ubtG0>

1. Introduction and scenario description

1.1 Background & Problem Statement

DHL Express handles millions of shipments globally, with operations teams constantly exchanging information through informal channels: MS Teams chat messages, email threads, screenshots, handwritten notes, and PowerPoint slides. While operationally critical, this knowledge remains unstructured and scattered, making it impossible to reuse efficiently. Creating clean, standardised Standard Operating Procedures (SOPs) or Knowledge Base articles is extremely slow and inconsistent.

1.2 Scenario Description

Scenario 1 requires designing and building a system that automates the complete transformation process from raw, messy input to clean, structured knowledge articles. The system must handle multiple input types, apply AI structuring, manage a review workflow, and integrate with RPA for automated ingestion.

1.3 Proposed Solution

The DHL Knowledge Base Management System is a full-stack web application integrated with UiPath RPA and Google Gemini AI. It delivers:

- Automated file ingestion from Google Drive using UiPath Studio
- AI-powered content structuring via Google Gemini 2.5 Flash (title, summary, steps, tags)
- Role-based access control: Admin, Editor, and Reviewer roles
- Full article lifecycle management: Draft → Reviewed → Published with version history
- Conflict detection flagging articles that overlap with existing published content
- Searchable, filterable knowledge base viewer with multi-criteria filtering
- Automated admin email notifications with run summaries after each RPA execution
- Searchable and filterable knowledge base viewer
- Summary email notifications to system administrators

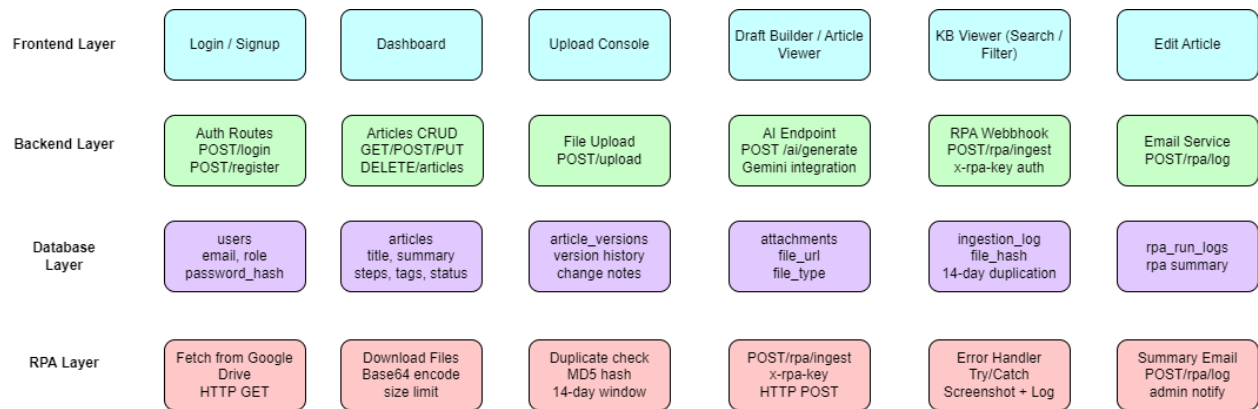
1.4 Logistics Context

In logistics, outdated or missing SOPs directly cause shipment errors, customs failures, and customer complaints. The DHL KB Automation System reduces the time from knowledge creation to publication from days to minutes, while ensuring quality through AI-assisted structuring and mandatory human review workflows before publication.

2. System architecture

2.1 Architecture Overview

The system follows a four-tier architecture: React frontend, Node.js/Express backend API, Supabase PostgreSQL database, and UiPath RPA automation layer. All tiers communicate via REST API over HTTP/HTTPS with JWT authentication.



Layer 1: Frontend (Presentation Layer)

The frontend is built using React 18 with Vite as the build tool. It provides a responsive single-page application (SPA) accessible through any modern web browser. The frontend handles all user interactions including login/signup, file uploads, AI-assisted draft creation, article review, and knowledge base searching.

Layer 2: Backend (Application Layer)

The backend is a RESTful API built with Node.js and Express.js. It serves as the central orchestrator for all business logic, including authentication, article management, file processing, AI integration, and RPA webhook handling.

Layer 3: Database (Data Layer)

The database layer uses Supabase, a cloud-hosted PostgreSQL database platform with built-in authentication, real-time subscriptions, and object storage. All data is persisted in a structured relational schema with proper foreign key relationships.

Layer 4: RPA Automation Layer

UiPath Studio is used to build the RPA workflow that automates the ingestion of files from Google Drive into the knowledge base system. The RPA bot runs on a scheduled or manual trigger basis and communicates with the backend API via HTTP requests.

2.2 Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite, React Router, Axios	SPA user interface, routing, API calls
Backend	Node.js + Express.js, JWT, bcryptjs	REST API, business logic, auth
Database	Supabase PostgreSQL + Storage	Data persistence, file storage
AI	Google Gemini 2.5 Flash	Content structuring, image OCR, conflict check

RPA	UiPath Studio, VB.NET Invoke Code	Automated file ingestion from Google Drive
Email	Nodemailer + Gmail SMTP	Admin summary notifications

2.3 Supabase Database Schema

The database consists of six core tables:

Table Name	Purpose
users	Stores user accounts with email, hashed password, full name, and role (admin/editor/reviewer)
articles	Main content table storing title, summary, steps (JSONB), tags (array), status, conflict flags, and creator references
article_versions	Immutable version history for every status change, capturing who changed what and when
attachments	Links uploaded files stored in Supabase Storage to their parent articles
ingestion_log	Records every RPA ingestion attempt with MD5 hash for 14-day duplicate detection
rpa_run_logs	Stores RPA execution summaries including counts of created, duplicated, and failed files

3. API and JSON structure

3.1 API Endpoints

All endpoints are prefixed with `/api/`. Protected routes require a JWT Bearer token. RPA routes use an `x-rpa-key` header instead.

Endpoint	Method	Description	Auth
<code>/api/auth/login</code>	POST	Authenticate, return JWT token	None
<code>/api/auth/register</code>	POST	Create new user account	None
<code>/api/articles</code>	GET	List articles (filter: status, tag, creator, search)	JWT
<code>/api/articles</code>	POST	Create new draft article	JWT
<code>/api/articles/:id</code>	PUT	Update article (role permissions enforced)	JWT
<code>/api/articles/:id</code>	DELETE	Delete article (admin or own draft only)	JWT
<code>/api/upload</code>	POST	Upload file, extract text content	JWT

/api/ai/generate	POST	Generate structured article from raw text/image	JWT
/api/rpa/ingest	POST	RPA file ingestion with AI processing	x-rpa-key
/api/rpa/log	POST	Save run summary, trigger admin email	x-rpa-key

3.2 JSON Examples

Article Object

```
{  "id": "a06155f7-...",  "title": "Resolving Label Printing Issues",  "summary": "Label failures commonly caused by spooler issues...",  "steps": ["Restart print spooler","Clear print queue","Reprint label"],  "tags": ["label","printing","troubleshooting"],  "status": "published",  "conflict_flag": false,  "created_by": "uuid",  "created_at": "2026-05-04T..." }
```

RPA Ingest Request

```
POST /api/rpa/ingest  Header: x-rpa-key: dhl-rpa-secret-2024
{  "fileName": "SOP.docx",  "fileContent": "base64...",  "mimeType": "application/vnd.openxmlformats-officedocument...",  "encoding": "base64",  "source": "google_drive" }
```

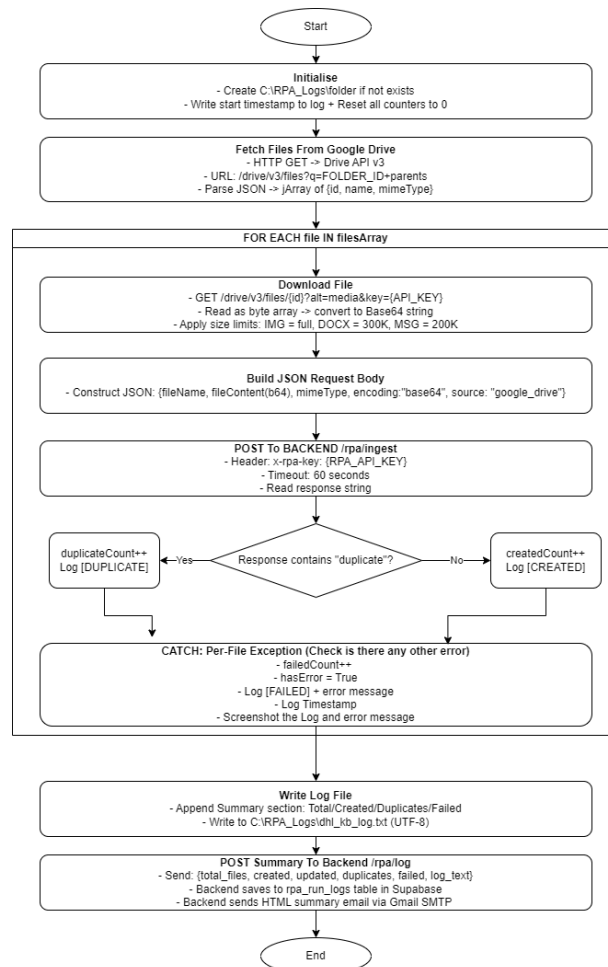
3.3 Role-Based Access Control

Action	Admin	Editor	Reviewer
Create Article	Yes	Yes	No
Edit Own Draft	Yes	Yes	No
Edit Any Article	Yes	No	No
Delete Own Draft	Yes	Yes	No
Delete Any Article	Yes	No	No
Mark as Reviewed	Yes	No	Yes
Publish Article	Yes	No	No
View All Articles	Yes	Yes	Yes

4. RPA workflow and automation explanation

4.1 Overview

The UiPath RPA workflow automates ingestion of files from a Google Drive folder into the DHL Knowledge Base. It monitors the folder, downloads files, processes them through the AI pipeline, and notifies the admin by email.



4.2 Backend AI Processing Pipeline

When /rpa/ingest receives a file, the backend executes this pipeline:

1. Decode Base64 content → extract readable text (DOCX: parse ASCII bytes; images: keep Base64 for Gemini Vision)
2. Sanitise text — remove non-ASCII characters that would corrupt JSON payloads
3. MD5 hash the file content → query ingestion_log for duplicates within the last 14 days; skip if found
4. Queue Gemini AI job — sequential processing prevents rate limit errors (one file at a time)
5. Auto-retry on 429 — read retryDelay from Gemini error response, wait exact duration + 10s buffer (max 3 attempts)

6. Check conflicts — require 2+ matching meaningful tags OR 3+ matching title words before flagging
7. Save article to Supabase with AI title, summary, steps, tags, conflict flag, and RPA Bot as creator
8. Upload original file to Supabase Storage, save attachment record linked to the article

4.3 Duplicate Detection Logic

Each file is hashed using MD5 and compared against the ingestion_log table for entries within the past 14 days. Duplicate files are skipped with a "duplicate" status logged. This prevents the same file from creating multiple articles when the RPA runs repeatedly on the same Drive folder.

5. User Interface Screenshots with Description

5.1 Login and Sign Up Page

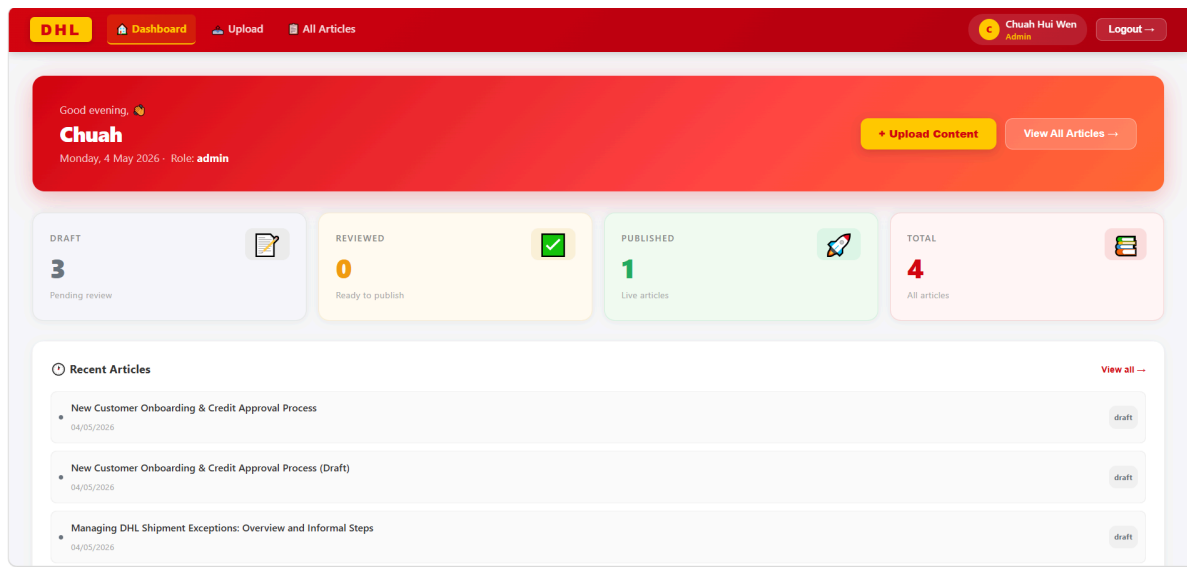
The screenshot shows the login interface of the DHL Knowledge Base System. At the top is a red gradient banner with the DHL logo and the text "KNOWLEDGE BASE SYSTEM". Below the banner are two tabs: "Login" (active) and "Sign Up". The login form includes an "Email Address" field with the placeholder "you@example.com", a "Password" field with a "Min. 6 characters" hint, and a red "Login --" button. At the bottom, there is a small footer: "DHL Logistics Operations - Internal Tool" and "© 2020 DHL Express. All rights reserved."

The screenshot shows the sign-up interface of the DHL Knowledge Base System. It features the same red gradient banner with the DHL logo and "KNOWLEDGE BASE SYSTEM" text. The "Sign Up" tab is active. The form includes fields for "Full Name" (placeholder: "e.g. Chua Hwen Wei"), "Email Address" (placeholder: "you@example.com"), and "Password" (placeholder: "Min. 6 characters"). There is also a "Role" dropdown menu currently set to "Admin — full access". A red "Create Account --" button is at the bottom.

The login page displays the DHL brand with a red gradient banner and yellow DHL logo. Users switch between Login and Sign Up via tabs. Sign Up collects full name, email, password, and

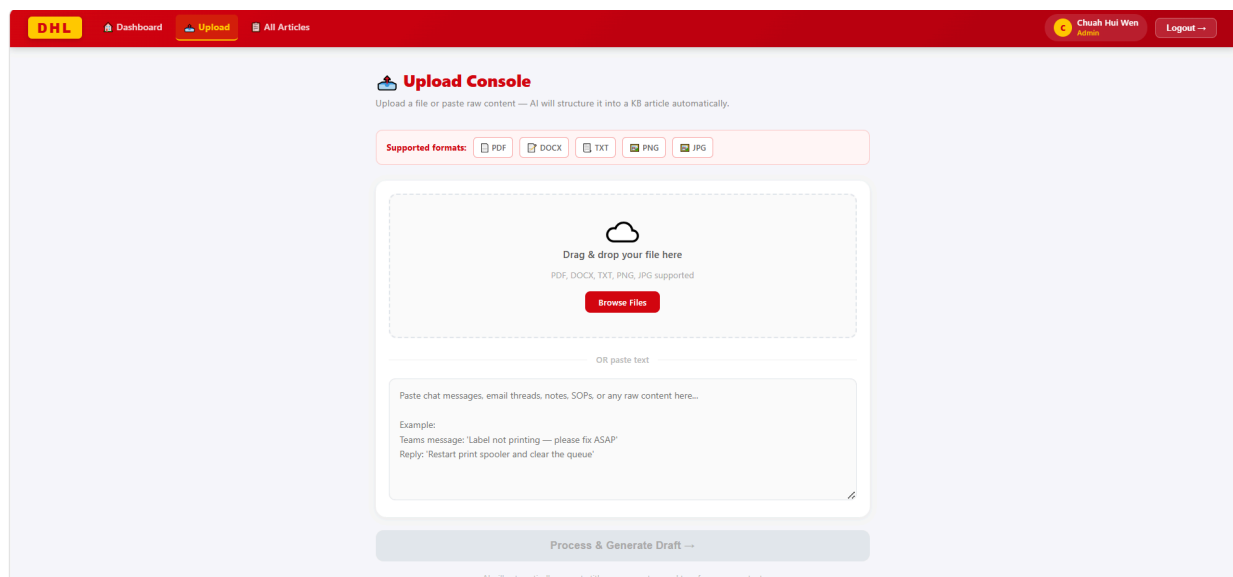
role (Editor/Reviewer/Admin). Input validation prevents submission of incomplete forms with inline error messages.

5.2 Dashboard



The login page displays the DHL brand with a red gradient banner and yellow DHL logo. Users switch between Login and Sign Up via tabs. Sign Up collects full name, email, password, and role (Editor/Reviewer/Admin). Input validation prevents submission of incomplete forms with inline error messages.

5.3 Upload Console



The login page displays the DHL brand with a red gradient banner and yellow DHL logo. Users switch between Login and Sign Up via tabs. Sign Up collects full name, email, password, and role (Editor/Reviewer/Admin). Input validation prevents submission of incomplete forms with inline error messages.

5.4 Edit Article Page

Edit Article Cancel

Role: admin

Title *

Booking Checklist Requirements

Summary

This message details the essential criteria for processing a booking. It specifies that a Customer PO, correct Incoterm, confirmed pickup date, and...

Steps

1. Customer PO ✕
2. Correct Incoterm ✕
3. Pickup date confirmed ✕
4. Weight & dimensions verified ✕

+ Add Step

Tags (comma separated)

booking, checklist, requirements, logistics, PO, incoterm

Save Cancel

The Edit Article page is only accessible to admins and editors for draft articles. All fields are fully editable including title, summary, individual steps, and comma-separated tags.

5.5 Knowledge Base Viewer

All Articles 3 articles found

Search by title...

STATUS TAG CREATOR

All Statuses All Tags All Creators

DATE FROM DATE TO

mm/dd/yyyy mm/dd/yyyy

Troubleshooting Label Printing Issues DRAFT

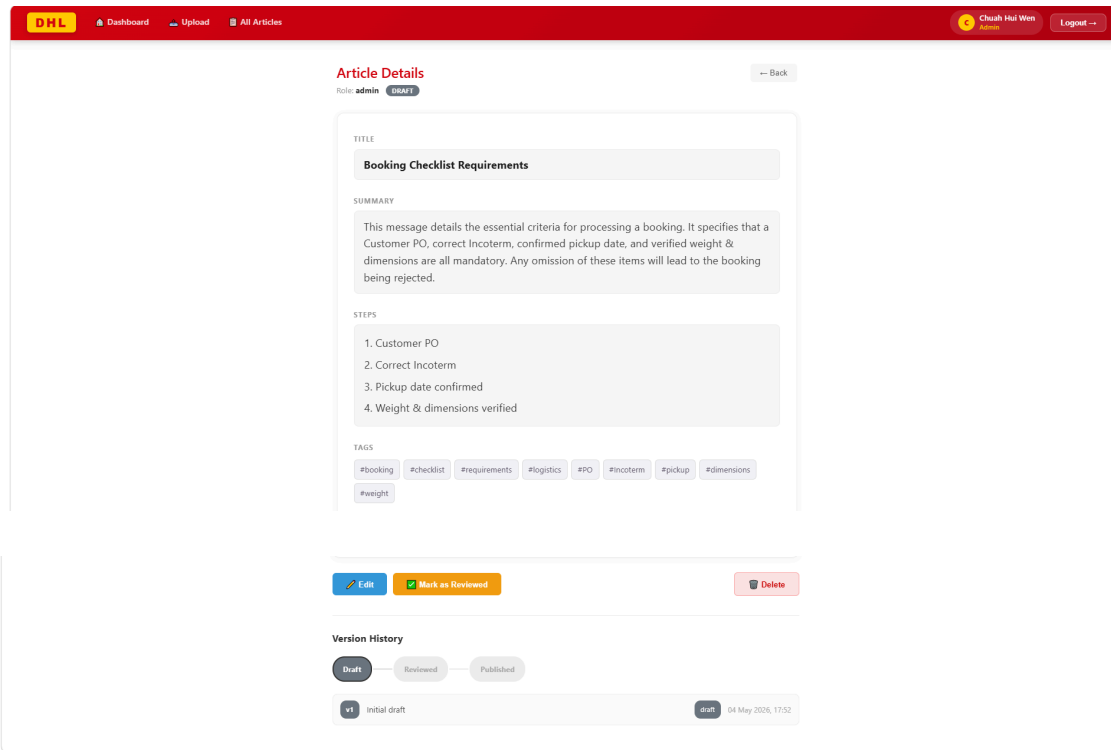
This article outlines steps to resolve common problems with labels not printing correctly or appearing blurry. It primar...

#printing #troubleshooting #printer #label #toner #cache #queue #DHL

Chuah Hui Wen 04 May 2025

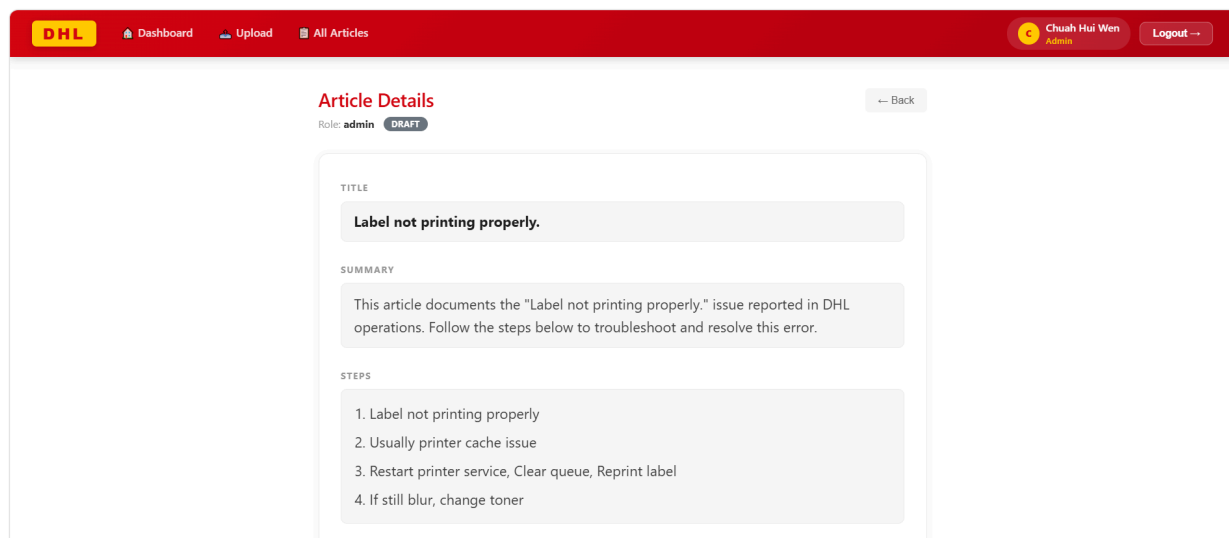
The KB Viewer provides a full-text search bar and four filter controls: Status, Tag (populated from existing tags), Creator (populated from article authors), and Date Range. Articles display as cards with a colour-coded left border by status, summary preview, tag chips, and creator/date metadata.

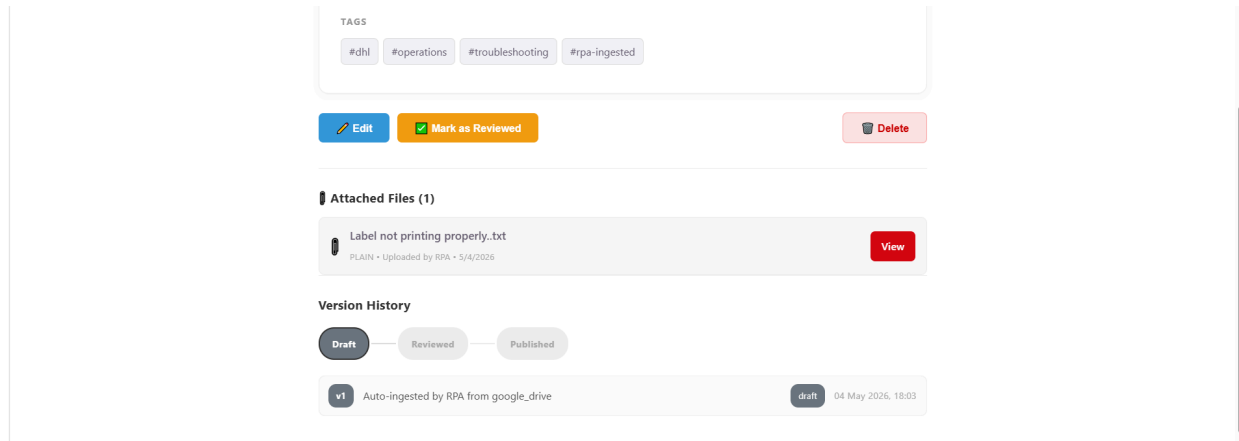
5.6 Draft Builder (Article Viewer)



The Article Viewer shows the complete structured article in read-only mode. Role-based action buttons (Edit, Mark as Reviewed, Publish, Delete) are conditionally shown. The Version History section displays a Draft→Reviewed→Published progress indicator and a chronological change log.

5.7 Attached Files





Articles ingested by RPA automatically have their original source files attached. The Attachments section displays each file with a type icon, file name, and upload date. A View button opens the file from Supabase Storage in a new tab, allowing reviewers to verify AI content against the original document.

6. Step-by-Step Instructions to run the system (user manual)

6.1 Prerequisites

Software	Requirement
Node.js	v18 or higher — nodejs.org
VS Code	Recommended IDE — code.visualstudio.com
UiPath Studio	Community edition — uipath.com
Web Browser	Google Chrome or Microsoft Edge (latest)

6.2 External Services Setup

Supabase

1. Create a project at supabase.com → Settings → API → copy Project URL and service_role key.
2. Run the full schema SQL in SQL Editor to create all 6 tables.
3. Go to Storage → New Bucket → name: attachments → toggle Public ON.

Google Gemini AI

4. Go to aistudio.google.com → Get API Key → Create API key in new project → copy key.

Gmail App Password

5. Enable 2-Step Verification on Google account → search App Passwords → generate for Mail.
6. Copy the 16-character password (no spaces) for use in .env as EMAIL_PASS.

Google Drive API

7. Go to console.cloud.google.com → Enable Google Drive API → Credentials → Create API Key.
8. Create a Drive folder, copy the folder ID from the URL for UiPath configuration.

6.3 Backend Setup

9. Open terminal in VS Code: `cd backend` → `npm install`
10. Create `.env` file in `backend/` with these values:

```
SUPABASE_URL=https://your-project.supabase.co
SUPABASE_SERVICE_KEY=your_service_role_key
JWT_SECRET=dhl_kb_secret_2026
GEMINI_API_KEY=AIzaSy_your_key
EMAIL_USER=your@gmail.com EMAIL_PASS=16charapppassword
ADMIN_EMAIL=admin@gmail.com RPA_API_KEY=dhl-rpa-secret-2024
RPA_USER_ID=uuid-from-supabase-rpa-user
```

11. Run in Supabase SQL Editor to create the RPA bot user:

```
INSERT INTO users (email,password_hash,full_name,role)
VALUES ('rpa@dhl.com','rpa-no-login','RPA Bot','editor');
SELECT id FROM users WHERE email='rpa@dhl.com'; -- copy UUID to .env
```

12. Start the backend: `node server.js` — verify at <http://localhost:5000>

6.4 Frontend Setup

13. Open second terminal: `cd frontend` → `npm install` → `npm run dev`
14. Open <http://localhost:5173> in browser.
15. Click Sign Up → enter details → select Admin role → Create Account → Login.

6.5 Manual Upload Workflow

16. Click Upload in navbar → drag a file or paste text → click Process & Generate Draft.
17. Review AI-generated title, summary, steps, and tags in the Article Viewer.
18. Click Edit to modify content if needed → Save.
19. Reviewer clicks Mark as Reviewed → Admin clicks Publish to make article live.

6.6 UiPath RPA Execution

20. Open UiPath Studio → open the RPA project → go to Main.xaml Variables panel.
21. Set: `backendUrl=http://localhost:5000/api`, `rpaApiKey=dhl-rpa-secret-2024`, `gDriveApiKey` and `gDriveFolderId` from Google setup.
22. Place test files into the Google Drive folder.
23. Press F5 to run → check `C:\RPA_Logs\dhl_kb_log.txt` for results.
24. Check admin email inbox for the HTML summary email.
25. Open <http://localhost:5173/viewer> to see newly created draft articles with attachments.